

The low carbon city

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y 2050, two-thirds of the world's population will be living in cities, up from the current 55%. This surge

towards urbanisation will be led by Asia, with India and China, along with Nigeria, accounting for 35% of the growth. What do these changes mean for our planet and how can scientists, like myself, help?

As an academic at the University of Edinburgh's School of Engineering, my job is to examine how we can use available resources as efficiently as possible, to reduce environmental impact. Sustainability is key to this transition. The growth in population is driving demand for more products and essential services like food, clean water, infrastructure, education, health, and housing. Such issues have risen to prominence, notably through the United Nations' Sustainable Development Goals (SDGs). However it is also important not to view urban areas in isolation. Hinterlands and rural environments supply vital resources. In improving a city, the country, and indeed the world, as a whole needs to be taken into account.

Climate change has the potential to catastrophically alter our environment and society. This phenomenon has implications for our society – we need to act radically and urgently to address it. Understanding what the low-carbon city is and how it can be rapidly achieved is key to approaching a sustainable society.

The low-carbon city requires that we improve the efficiency in how we use materials and energy. This includes the provision of lighting, water, food, and consumer products that we use in our daily lives, as well as low CO₂ emission transport systems and buildings. This must also involve environmentally beneficial processing of waste/by-product materials that have the potential to replace primary materials.

Developing solutions by working in partnership is a key approach for the University of Edinburgh. One such collabora-



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tion is with colleagues at IIT Madras, with whom I am developing alternatives to the use of conventional Portland cement. This material is responsible for around 6% of worldwide CO₂ emissions, and without transformative changes to the sector, could contribute up to one third of all CO₂ emissions in 2050. My IIT partners and I are looking at using novel lower environmental impact materials to replace cement, such as incinerator ash. The cost and environmental savings, both in reusing wastes and by avoiding primary materials extraction and processing – such as mining for raw materials – could be enormous. Our initial work has been promising – the next stage is to analyse our alternative cement materials on the mass scale.

With 250 million people set to join the urban population of India by 2030, the requirement for more buildings, and more cement is obvious. Of course, any

attempt to change current practices needs the support of the population. Persuading people that a proposed alteration that could impact and change their lives in a significant way is not always straight-forward. One example of this, and one currently being debated in Edinburgh, concerns the role and use of vehicles in cities. Scotland's capital city is an architectural gem. Often voted UK's best place to live, it contains outstanding buildings spanning centuries of development. What it is not suited to it are the current levels of traffic. Private cars, taxis, lorries, and buses clog the city's streets and air pollution is an increasing concern. The low-carbon city of the future will inevitably see very different transport systems. But with competing demands from those who live in the city, those who are visiting as tourists, and those who rely on a private vehicle for business, Edinburgh's city council has a big job in identifying and persuading all of these key stakeholders that viable alternative systems could work. It is almost inevitable that the solution will be a complex mix of various technologies. Such complexity, and trade-offs between different technologies, are essential to the sustainable society and low-carbon city. However, we need to realise these alternative futures if we are to achieve a sustainable society.

This, and many other questions, will be discussed next month at a conference in China, organised jointly by the universities of Edinburgh and Shanghai Jiao Tong, and the British Council. China, already home to countless 'mega cities', is a highly appropriate venue for these workshops, which will see a broad range of academic experts coming together. With social scientists joining engineers, physicists, chemists, and industrial ecologists like myself, the hope is that this broad range of knowledge will provide new approaches to the varied questions surrounding our low-carbon future. But one thing is clear – the status quo is not an option. **d**